Common Design Problems of Attributes

Identify data that can be calculated

If you know	You can calculate
Game teams + scores	Winning team
Who manages whom	Number of subordinates
Student scores	Letter grades
Racer times	Race ranking
Item shipped date	Has item shipped?
Purchased item prices + quantities	Purchase subtotal
Purchased item prices + quantities + location + date	Purchase total, including taxes

Avoid data that needs regular updates

Data that needs to be updated	Unchanging data
Age	Birth date
Duration of employment	Hire date
Baseball batting average	Result of each at-bat or each pitch
Total commissions	Individual sales + commission rate

Group/ungroup data into attributes

Data	Common incorrect solution	Correct solution	Comments
Game score	One string like ‘47-14’	Two numbers, one for each team	Might want to aggregate a team’s scores
Height	Two numbers (6, 4) or one string ‘6 foot 4’	One number (76)	A single measurement
Address	One string	Separate strings for city, state. Probably no need to separate house number from street name except for logistics company planning delivery routes.	Depends on how organization will use data.
Name	One string	Two or more strings	Often need the individual pieces
Days of the week worked	One string like ‘MWF’	Put employees and days in a many-many relationship	Don’t encode multiple values into one.
Period worked	One string like ‘2-4pm’	Starting time plus either duration or ending time	Don’t encode multiple values into one.